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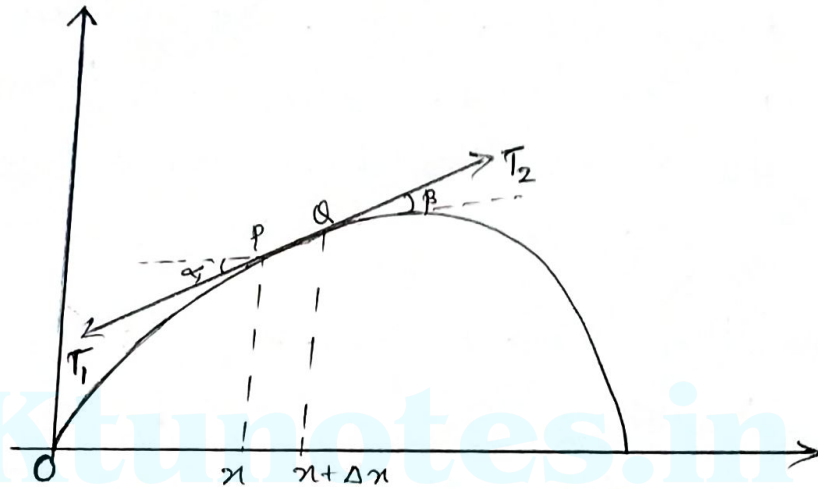
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MODULE - 2

Applications of Partial Differential Equations

One Dimensional Wave Equation:



consider a tightly stretched string of length l with its end points fixed. Let the string be released from rest and allowed to vibrate. The problem is to find the deflection $y(x, t)$ at any point x and at any time $t \geq 0$. For deriving the partial differential equation we make the following assumptions:

- i) The mass of the string per unit length is constant.
- ii) The string is perfectly elastic and does not offer any resistance to bending.
- iii) The tension caused by stretching the string before fixing it at the endpoints is so large compared to the weight of the string and hence the action of the gravitational force can be neglected.
- iv) the string performs a small transverse vibration in a vertical plane. In other words, every particle of the string moves strictly vertically so that the deflection and the slope at every point of the string remains small in absolute value.

Take one end of the string as origin O and take the x -axis along the string. We assume that the motion takes place entirely in the xy -plane. Consider the forces acting on a small portion PQ of the string where $P(x, y)$ and $Q(x+\Delta x, y+\Delta y)$. Since the string does not offer resistance to bending, the tension at each point

of the string is tangential to the curve of the string. Let T_1 and T_2 be the tensions at the points P and Q. Let α and β be the angles made by the tangents at P and Q respectively with the x-axis. Since the points of the string move vertically there is no motion in the horizontal direction. Hence the horizontal components of the tension must be constant.

$$\therefore T_1 \cos \alpha = T_2 \cos \beta = T (\text{constant})$$

Let m be the mass per unit length of the string. The vertical component of the force acting on the element PQ is

$$T_2 \sin \beta - T_1 \sin \alpha$$

Acceleration of the element in the y-direction is $\frac{\partial^2 y}{\partial t^2}$

By Newton's second law of motion,

$$F = Ma$$

$$\therefore T_2 \sin \beta - T_1 \sin \alpha = m \Delta x \frac{\partial^2 y}{\partial t^2}$$

Divide by $T_1 \cos \alpha = T_2 \cos \beta = T$ we get,

$$\frac{T_2 \sin \beta}{T_2 \cos \beta} - \frac{T_1 \sin \alpha}{T_1 \cos \alpha} = \frac{m \Delta x}{T} \frac{\partial^2 y}{\partial t^2}$$

$$\Rightarrow \tan \beta - \tan \alpha = \frac{m \Delta x}{T} \frac{\partial^2 y}{\partial t^2}$$

$\tan \alpha$ and $\tan \beta$ are the slopes of the string at P and Q

$$\therefore \tan \alpha = \left(\frac{\partial y}{\partial x} \right)_P \text{ and } \tan \beta = \left(\frac{\partial y}{\partial x} \right)_Q$$

$$\left(\frac{\partial y}{\partial x} \right)_Q - \left(\frac{\partial y}{\partial x} \right)_P = \frac{m \Delta x}{T} \frac{\partial^2 y}{\partial t^2}$$

$$\Rightarrow \frac{\partial^2 y}{\partial t^2} = \frac{T}{m} \left[\frac{\left(\frac{\partial y}{\partial x} \right)_Q - \left(\frac{\partial y}{\partial x} \right)_P}{\Delta x} \right]$$

$$\frac{\partial^2 y}{\partial t^2} = \frac{T}{m} \frac{\partial^2 y}{\partial x^2}$$

$$\therefore \boxed{\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2}}; \quad a^2 = \frac{T}{m}$$

This pde is known as one dimensional wave equation.

Solution of one dimensional wave equation by method of separation of variables.

One dimensional wave equation is

$$\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2} \longrightarrow \textcircled{1}$$

Let $y(x, t) = x(x) T(t)$, where x is a function of x and T is a function of t only, be the solution of $\textcircled{1}$.

$$\text{Then } \frac{\partial^2 y}{\partial t^2} = x T'' \quad ; \quad \frac{\partial^2 y}{\partial x^2} = x'' T$$

$$\therefore \textcircled{1} \Rightarrow x T'' = a^2 x'' T$$

$$\Rightarrow \frac{T''}{T} = a^2 \frac{x''}{x}$$

$$\text{or } \frac{x''}{x} = \frac{1}{a^2} \frac{T''}{T}$$

clearly LHS is a function of x alone and RHS is a function of t alone.

$$\therefore \frac{x''}{x} = k \quad \text{and} \quad \frac{1}{a^2} \frac{T''}{T} = k.$$

$$\Rightarrow \frac{d^2 x}{dx^2} = k x \quad ; \quad \frac{1}{a^2} \frac{d^2 T}{dt^2} = k T$$

The solution of the above equation depends on the value of k .

case (i) : when k is positive, i.e. let $k = \lambda^2$

$$\frac{d^2 x}{dx^2} = \lambda^2 x \quad ; \quad \frac{d^2 T}{dt^2} = \lambda^2 a^2 T$$

$$\Rightarrow \frac{d^2 x}{dx^2} - \lambda^2 x = 0 \quad ; \quad \frac{d^2 T}{dt^2} - \lambda^2 a^2 T = 0$$

$$\Rightarrow (D^2 - \lambda^2) x = 0 \quad ; \quad (D^2 - a^2 \lambda^2) T = 0$$

$$\text{A.E. is } m^2 - \lambda^2 = 0$$

$$\Rightarrow m = \pm \lambda$$

$$\therefore x = c_1 e^{\lambda x} + c_2 e^{-\lambda x}$$

$$\text{A.E. is } m^2 - a^2 \lambda^2 = 0$$

$$\Rightarrow m = \pm a \lambda$$

$$T = c_3 e^{a \lambda t} + c_4 e^{-a \lambda t}$$

$$\therefore u(x, t) = (c_1 e^{\lambda x} + c_2 e^{-\lambda x}) (c_3 e^{a \lambda t} + c_4 e^{-a \lambda t})$$

case (ii) when k is negative, let $k = -\lambda^2$

$$\frac{d^2x}{dx^2} + \lambda^2 x = 0 \quad ; \quad \frac{d^2T}{dt^2} + a^2 \lambda^2 T = 0$$

$$\Rightarrow (D^2 + \lambda^2)x = 0$$

$$(D^2 + a^2 \lambda^2)T = 0$$

$$\text{A.E. is } m^2 + \lambda^2 = 0$$

$$\text{A.E. is } m^2 + a^2 \lambda^2 = 0$$

$$m = \pm i \lambda$$

$$m = \pm i a \lambda$$

$$x = c_5 \cos \lambda x + c_6 \sin \lambda x \quad ; \quad T = c_7 \cos a \lambda t + c_8 \sin a \lambda t$$

$$\therefore y(x, t) = (c_5 \cos \lambda x + c_6 \sin \lambda x) (c_7 \cos a \lambda t + c_8 \sin a \lambda t)$$

case (iii) : when $k = 0$.

$$\frac{d^2x}{dx^2} = 0 \quad ; \quad \frac{d^2T}{dt^2} = 0$$

$$\Rightarrow \frac{dx}{dx} = c_9$$

$$\frac{dT}{dt} = c_{11}$$

$$\Rightarrow x = c_9 x + c_{10}$$

$$T = c_{11} t + c_{12}$$

$$\therefore y(x, t) = (c_9 x + c_{10}) (c_{11} t + c_{12})$$

Thus the possible solutions of the wave equation are

$$y(x, t) = (c_1 e^{\lambda x} + c_2 e^{-\lambda x}) (c_3 e^{a \lambda t} + c_4 e^{-a \lambda t})$$

$$y(x, t) = (c_5 \cos \lambda x + c_6 \sin \lambda x) (c_7 \cos a \lambda t + c_8 \sin a \lambda t)$$

$$y(x, t) = (c_9 x + c_{10}) (c_{11} t + c_{12})$$

As we are dealing with the problem on vibrations, y must be periodic function of x and t . Hence their solution must involve trigonometric terms. Thus the only suitable solution of the wave equation is

$$y(x, t) = (A \cos \lambda x + B \sin \lambda x) (C \cos a \lambda t + D \sin a \lambda t)$$

The form of the solution, that of the one dimensional wave equation $\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2}$ and the physical nature of the problem shows that we need four conditions to find a unique solution. The term $\frac{\partial^2 y}{\partial t^2}$ indicates that we need two initial conditions and $\frac{\partial^2 y}{\partial x^2}$ shows we need two boundary conditions.

The two boundary conditions are:

$$i) y(0, t) = a(t) \quad ii) y(l, t) = b(t), \quad \forall t \geq 0$$

where $a(t)$ and $b(t)$ are either constants or functions of t alone. The initial conditions are

$$iii) y(x, 0) = f(x), \quad 0 \leq x \leq l$$

$$iv) \left(\frac{\partial y}{\partial t} \right)_{t=0} = g(x), \quad 0 \leq x \leq l$$

where $f(x)$ is called the initial deflection of the string and $g(x)$ is called initial velocity given to each points on the string

Problems:-

- Q. A string is stretched and fastened to two points l apart. Motion is started by displacing the string in the form, $y = a \sin(\frac{\pi x}{l})$ from which it is released at time $t = 0$. Show that the displacement of any point at a distance x from one end at time t is given by $y(x, t) = a \sin(\frac{\pi x}{l}) \cos(\frac{\pi a t}{l})$.

Soln:

The vibration of the string is given by

$$\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2} \longrightarrow \textcircled{1}$$

As the end points of the string are fixed, for all time,

$$y(0, t) = 0, \quad y(l, t) = 0.$$

Since the initial transverse velocity of any point of the string is zero,

$$\therefore \left(\frac{\partial y}{\partial t} \right)_{t=0} = 0.$$

$$\text{Also } y(x, 0) = a \sin\left(\frac{\pi x}{l}\right)$$

Now we have to solve $\textcircled{1}$ subject to the boundary conditions and initial conditions

$\therefore$ The solution is of the form,

$$y(x,t) = (A \cos \lambda x + B \sin \lambda x)(C \cos \lambda ct + D \sin \lambda ct) \rightarrow (2)$$

$$y(0,t) = 0 \Rightarrow$$

$$0 = A(C \cos \lambda ct + D \sin \lambda ct)$$

$$\Rightarrow A = 0.$$

$$\therefore (2) \rightarrow y(x,t) = B \sin \lambda x (C \cos \lambda ct + D \sin \lambda ct) \rightarrow (3)$$

$$\frac{\partial y}{\partial t} = B \sin \lambda x [C(-c\lambda \sin \lambda ct) + D(c\lambda \cos \lambda ct)]$$

$$\left(\frac{\partial y}{\partial t}\right)_{t=0} = B \sin \lambda x (Dc\lambda) = 0.$$

If $B = 0$ we get only the trivial solution.

$$y(x,t) = 0.$$

$\therefore$ the only possibility is $D = 0$.

$$\text{Thus } (3) \Rightarrow y(x,t) = B \sin \lambda x (C \cos \lambda ct) \rightarrow (4).$$

By $y(L,t) = 0$,

the above becomes,

$$0 = B \sin \lambda L (C \cos \lambda ct)$$

$$\text{Since } B, C \neq 0 \Rightarrow \sin \lambda L = 0$$

$$\Rightarrow \lambda L = n\pi$$

$$\Rightarrow \lambda = \frac{n\pi}{L} \text{ where } n \text{ is an integer}$$

$$\text{Hence } (4) \Rightarrow y(x,t) = B \sin\left(\frac{n\pi x}{L}\right) C \cos\left(\frac{n\pi ct}{L}\right).$$

Finally applying the last condition,

$$y(x,0) = a \sin\left(\frac{n\pi x}{L}\right), \text{ we get}$$

a $\sin \frac{\pi n}{l} = BC \sin \left(\frac{n\pi x}{l} \right)$, which will be satisfied by taking $B=C=a$ and $n=1$.

Hence the required solution is

$$\underline{\underline{y(x,t) = a \sin \left(\frac{\pi x}{l} \right) \cos \left(\frac{\pi ct}{l} \right)}}$$

- Q. A tightly stretched string with fixed end points $x=0$ and $x=l$ is initially in a position given by $y = y_0 \sin^3 \left(\frac{\pi x}{l} \right)$. If it is released from rest from this position, find the displacement $y(x,t)$.

Solution

The equation of the vibrating string is

$$\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2} \quad \longrightarrow \textcircled{1}$$

The boundary conditions are

$$y(0,t) = 0, \quad y(l,t) = 0.$$

Also the initial conditions are

$$y(x,0) = y_0 \sin^3 \frac{\pi x}{l}$$

$$\left(\frac{\partial y}{\partial t} \right)_{t=0} = 0.$$

The solution of $\textcircled{1}$ is of the form,

$$y(x,t) = (A \cos \lambda x + B \sin \lambda x) (C \cos \lambda at + D \sin \lambda at) \quad \longrightarrow \textcircled{2}$$

Applying the boundary conditions, we get
 $0 = A (c \cos \lambda at + B \sin \lambda at)$

$$\Rightarrow A = 0$$

$$\textcircled{2} \Rightarrow \therefore y(x, t) = (B \sin \lambda x) (c \cos \lambda at + B \sin \lambda at) \rightarrow \textcircled{3}$$

$$\text{Now } y(l, t) = 0 \Rightarrow$$

$$0 = B \sin \lambda l (c \cos \lambda at + B \sin \lambda at)$$

$$\Rightarrow B \sin \lambda l = 0 \quad (B \text{ cannot be equal to zero})$$

$$\Rightarrow \sin \lambda l = 0$$

$$\Rightarrow \lambda l = n\pi$$

$$\Rightarrow \lambda = \frac{n\pi}{l}$$

$$\textcircled{3} \Rightarrow \therefore y(x, t) = B \sin\left(\frac{n\pi x}{l}\right) \left(c \cos \frac{n\pi at}{l} + B \sin \frac{n\pi at}{l}\right) \rightarrow \textcircled{4}$$

$$\frac{\partial y}{\partial t} = \left[B \sin\left(\frac{n\pi x}{l}\right) \right] \left[-c \sin\left(\frac{n\pi at}{l}\right) \frac{n\pi a}{l} + B \cos\left(\frac{n\pi at}{l}\right) \frac{n\pi a}{l} \right]$$

$$\left[\frac{\partial y}{\partial t} \right]_{t=0} = 0 \Rightarrow 0 = \left[B \sin\left(\frac{n\pi x}{l}\right) \right] D \left(\frac{n\pi a}{l}\right).$$

$$\text{i.e., } D = 0.$$

$$\textcircled{4} \Rightarrow y(x, t) = B \sin\left(\frac{n\pi x}{l}\right) c \cos\left(\frac{n\pi at}{l}\right) \\ = A_n \sin\left(\frac{n\pi x}{l}\right) \cos\left(\frac{n\pi at}{l}\right)$$

Adding all such solutions the general solution of $\textcircled{1}$ is

$$y(x,t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{l}\right) \cos\left(\frac{n\pi at}{l}\right) \longrightarrow \textcircled{5}$$

$\therefore$ from $y(x,0) = y_0 \sin^3\left(\frac{\pi x}{l}\right)$, we have

$$y_0 \sin^3 \frac{\pi x}{l} = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{l}\right).$$

$$\text{or } y_0 \left\{ \frac{3 \sin\left(\frac{\pi x}{l}\right) - \sin\left(\frac{3\pi x}{l}\right)}{4} \right\} = A_1 \sin \frac{\pi x}{l} + A_2 \sin\left(\frac{2\pi x}{l}\right) + A_3 \sin\left(\frac{3\pi x}{l}\right) + \dots$$

comparing both sides, we have

$$A_1 = \frac{3y_0}{4}, A_2 = 0, A_3 = -\frac{y_0}{4}, A_4 = A_5 = A_6 = \dots = 0.$$

Hence from $\textcircled{5}$, the solution is,

$$y(x,t) = \frac{3y_0}{4} \sin \frac{\pi x}{l} \cos \frac{\pi at}{l} - \frac{y_0}{4} \sin \frac{3\pi x}{l} \cos \frac{3\pi at}{l}.$$

Q. A tightly stretched string with fixed endpoints $x=0$ and $x=l$ is initially at rest in its equilibrium position. If it is vibrating by giving to each of its points a velocity $\lambda x(l-x)$, find the displacement of the string at any distance x from one end at any time t .

Solution

The equation the vibrating string is

$$\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2} \longrightarrow \textcircled{1}$$

The boundary conditions are,

$$i) y(0, t) = 0, \quad ii) y(l, t) = 0$$

Also the initial conditions are

$$iii) y(x, 0) = 0, \quad iv) \left(\frac{\partial y}{\partial t} \right)_{t=0} = \lambda x(l-x)$$

The general solution of ① satisfying the boundary conditions and $y(x, 0) = 0$ is

$$y(x, t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{l}\right) \sin\left(\frac{n\pi a t}{l}\right) \rightarrow ②$$

$$\frac{\partial y}{\partial t} = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{l}\right) \cos\left(\frac{n\pi a t}{l}\right) \left(\frac{n\pi a}{l}\right)$$

$$\left(\frac{\partial y}{\partial t} \right)_{t=0} = \lambda x(l-x) \text{ implies,}$$

$$\lambda x(l-x) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{l}\right) \left(\frac{n\pi a}{l}\right)$$

$$\therefore \lambda x(l-x) = \frac{\pi a}{l} \sum_{n=1}^{\infty} n A_n \sin\left(\frac{n\pi x}{l}\right)$$

$$\therefore \frac{\pi a}{l} n A_n = \frac{2}{l} \int_0^l \lambda x(l-x) \sin\left(\frac{n\pi x}{l}\right) dx.$$

$$= \frac{2\lambda}{l} \left[(lx - x^2) \left(-\frac{1}{n\pi} \cos \frac{n\pi x}{l} \right) - (l - 2x) \left(-\frac{l^2}{n^2 \pi^2} \sin \frac{n\pi x}{l} \right) + (-2) \frac{l^3}{n^3 \pi^3} \cos \frac{n\pi x}{l} \right]_0^l$$

$$= \frac{4\lambda l^2}{n^3 \pi^3} (1 - \cos n\pi) = \frac{4\lambda l^2}{n^3 \pi^3} [1 - (-1)^n]$$

$$A_n = \frac{4\lambda l^3}{a \pi^4} [1 - (-1)^n].$$

Hence the desired solution is,

$$y(x,t) = \frac{4\lambda l^3}{a\pi^4} \left[\sum_{n=1}^{\infty} (1 - (-1)^n) \sin \frac{n\pi x}{l} \sin \left(\frac{n\pi a t}{l} \right) \right]$$

D'Alembert's solution of the wave equation.

wave equation is $\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2}$. $\longrightarrow \textcircled{1}$

Let us introduce the new independent variables $u = x+at$, $v = x-at$ so that y becomes a function of u and v .

Then $\frac{\partial y}{\partial x} = \frac{\partial y}{\partial u} + \frac{\partial y}{\partial v}$ and

$$\begin{aligned} \frac{\partial^2 y}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial y}{\partial u} + \frac{\partial y}{\partial v} \right) \\ &= \frac{\partial}{\partial u} \left(\frac{\partial y}{\partial u} + \frac{\partial y}{\partial v} \right) + \frac{\partial}{\partial v} \left(\frac{\partial y}{\partial u} + \frac{\partial y}{\partial v} \right) \\ &= \frac{\partial^2 y}{\partial u^2} + 2 \frac{\partial^2 y}{\partial u \partial v} + \frac{\partial^2 y}{\partial v^2} \end{aligned}$$

Similarly $\frac{\partial^2 y}{\partial t^2} = a^2 \left(\frac{\partial^2 y}{\partial u^2} - 2 \frac{\partial^2 y}{\partial u \partial v} + \frac{\partial^2 y}{\partial v^2} \right)$

Substituting the values of $\frac{\partial^2 y}{\partial x^2}$ and $\frac{\partial^2 y}{\partial t^2}$ in $\textcircled{1}$, we get

$$\frac{\partial^2 y}{\partial u \partial v} = 0 \longrightarrow \textcircled{2}$$

Integrating $\textcircled{2}$ w.r.t. v , we get $\frac{\partial y}{\partial u} = f(u)$; where $f(u)$ is an arbitrary function of u .

Now integrating $\frac{\partial y}{\partial u} = f(u)$ w.r.t. u , we get

$$y = \int f(u) du + \psi(v)$$

Since the integral is a function of u alone,

$$y = \phi(u) + \psi(v)$$

$$y(x, t) = \phi(x+ct) + \psi(x-ct) \longrightarrow (3)$$

This is the general solution of the wave equation

Now to determine ϕ and ψ , suppose initially,

$$u(x, 0) = f(x) \text{ and } \left[\frac{\partial y}{\partial t} \right]_{(x, 0)} = 0.$$

Differentiating (3) w.r.t. t , we get,

$$\frac{\partial y}{\partial t} = c \phi'(x+ct) - c \psi'(x-ct)$$

$$\text{At } t=0, \phi'(x) = \psi'(x) \text{ and } (4)$$

$$y(x, 0) = \phi(x) + \psi(x) = f(x) \longrightarrow (5)$$

$$(4) \text{ gives, } \phi(x) = \psi(x) + k$$

$$\therefore (5) \text{ becomes, } 2\psi(x) + k = f(x)$$

$$\psi(x) = \frac{1}{2} [f(x) - k]$$

$$\text{and } \phi(x) = \frac{1}{2} [f(x) + k]$$

Hence the solution of (3) takes the form

$$y(x, t) = \frac{1}{2} [f(x+ct) + k] + \frac{1}{2} [f(x-ct) - k]$$

$$= \frac{1}{2} [f(x+ct) + f(x-ct)]$$

$$y(x, t) = \frac{1}{2} [f(x+ct) + f(x-ct)]$$

which is the D'Alembert's solution of the wave equation.

Q. Find the deflection of a vibrating string of unit length having fixed ends with initial velocity zero and initial deflection $f(x) = K(\sin x - \sin 2x)$.

Soln:

By D'Alembert's method, the solution is $y(x, t) = \frac{1}{2} [f(x+ct) + f(x-ct)]$

$$\begin{aligned} &= \frac{1}{2} [K(\sin(x+ct) - \sin 2(x+ct)) \\ &\quad + K(\sin(x-ct) - \sin 2(x-ct))] \\ &= K[\sin x \cos ct - \sin 2x \cos 2ct] \end{aligned}$$

$$\text{Also, } y(x, 0) = K(\sin x - \sin 2x) = f(x)$$

$$\left[\frac{\partial y}{\partial t} \right]_{(x,0)} = K[-c \sin x \sin ct + 2c \sin 2x \sin 2ct] = 0$$

i.e., the given boundary conditions are satisfied. =====

One Dimensional Heat equation

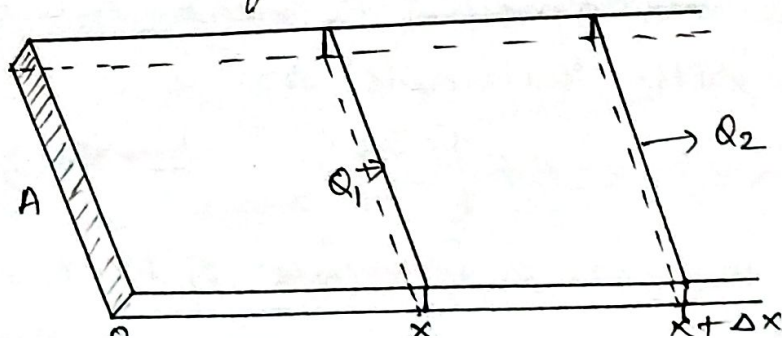
The one dimensional heat equation is derived on the basis of following empirical laws:

- i) Heat flows from higher temperature to lower temperature.
- ii) The amount of heat crossing any section of the bar per second is directly proportional to the area A of the cross section and the temperature gradient.

The proportionality constant k is called the thermal conductivity of the material.

- iii) The quantity of heat required to produce a given temperature change in a body is proportional to the mass of the body and to temperature change.

The proportionality constant c is called the specific heat capacity of the material.



considers the flow of heat of uniform temperature in a uniform bar. It is assumed that the sides of the bar are insulated and the loss of heat from the sides by conduction or radiation is negligible. Take one end of the bar as origin and the direction of flow as the positive x -axis. The temperature u at any point of the bar depends on the distance x of the point from one end and the time t . Also the temperature of all points of any cross section is the same.

Let Q_1 be the quantity of heat flowing into the section at a distance x . Then,

$$Q_1 = -KA \left(\frac{\partial u}{\partial x} \right)_x \text{ per section}$$

(-ve sign is attached because as x increases u decreases)

Let Q_2 be the quantity of heat flowing out of the section at a distance $x + \Delta x$.

$$\text{Then } Q_2 = -KA \left(\frac{\partial u}{\partial x} \right)_{x+\Delta x} \text{ per section.}$$

Hence the amount of heat retained by the slab with thickness Δx is

$$Q_1 - Q_2 = KA \left[\left(\frac{\partial u}{\partial x} \right)_{x+\Delta x} - \left(\frac{\partial u}{\partial x} \right)_x \right]$$

But the rate of increase of heat in the slab

is $c \rho A \Delta x \frac{\partial y}{\partial t}$ where c is the specific heat capacity of the material and ρ , the density of the material.

$$\therefore c \rho A \Delta x \frac{\partial y}{\partial t} = kA \left[\left(\frac{\partial y}{\partial x} \right)_{x+\Delta x} - \left(\frac{\partial y}{\partial x} \right)_x \right]$$

$$\therefore \frac{\partial y}{\partial t} = \frac{k}{c \rho} \left[\frac{\left(\frac{\partial y}{\partial x} \right)_{x+\Delta x} - \left(\frac{\partial y}{\partial x} \right)_x}{\Delta x} \right]$$

Taking the limit as $\Delta x \rightarrow 0$, we have

$$\frac{\partial y}{\partial t} = \frac{k}{c \rho} \frac{\partial^2 y}{\partial x^2}$$

$$\boxed{\frac{\partial y}{\partial t} = \alpha^2 \frac{\partial^2 y}{\partial x^2}} ; \text{ where } \alpha^2 = \frac{k}{c \rho} \text{ is known as the diffusivity of the material of bar.}$$

Solution of one dimensional heat equation (using method of separation of variables)

one dimensional heat equation is

$$\frac{\partial y}{\partial t} = \alpha^2 \frac{\partial^2 y}{\partial x^2} \longrightarrow \textcircled{1}$$

Let $u(x, t) = X(x)T(t)$, where x is a function of x and T is a function of t alone.

$$\frac{\partial y}{\partial t} = XT' \quad \text{and} \quad \frac{\partial^2 y}{\partial x^2} = X''T$$

$$\therefore \textcircled{1} \Rightarrow XT' = \alpha^2 X''T$$

$$\Rightarrow \underline{X''} = \frac{1}{\alpha^2} \underline{T'} \longrightarrow \textcircled{2}$$

clearly LHS of (2) is a function of x alone and RHS is a function of t only.

$$\therefore \frac{x''}{x} = k \quad \text{and} \quad \frac{1}{\alpha^2} \frac{T'}{T} = k.$$

$$\Rightarrow \frac{d^2 x}{dx^2} = kx \quad ; \quad \frac{dT}{dt} = k\alpha^2 T$$

$$\Rightarrow \frac{d^2 x}{dx^2} - kx = 0 \quad \frac{dT}{dt} - k\alpha^2 T = 0$$

The solution of the above equations depends on the value of k .

case (i): when k is positive ($k = \lambda^2$)

$$\frac{d^2 x}{dx^2} - \lambda^2 x = 0 \quad ; \quad \frac{dT}{dt} - \lambda^2 \alpha^2 T = 0$$

$$\Rightarrow (D^2 - \lambda^2) x = 0$$

$$(D - \lambda^2 \alpha^2) T = 0$$

$$\text{A.E. is } m^2 - \lambda^2 = 0$$

$$\text{A.E. is } m - \lambda^2 \alpha^2 = 0$$

$$\Rightarrow m = \pm \lambda$$

$$\Rightarrow m = \lambda^2 \alpha^2$$

$$x = C_1 e^{\lambda x} + C_2 e^{-\lambda x}$$

$$T = C_3 e^{\lambda^2 \alpha^2 t}$$

case (ii): when k is negative ($k = -\lambda^2$)

$$\frac{d^2 x}{dx^2} + \lambda^2 x = 0 \quad ; \quad \frac{dT}{dt} + \lambda^2 \alpha^2 T = 0$$

$$(D + \lambda^2 \alpha^2) T = 0$$

$$(D^2 + \lambda^2) x = 0$$

$$\text{A.E. is } m + \lambda^2 \alpha^2 = 0$$

$$\text{A.E. is } m^2 + \lambda^2 = 0$$

$$m = -\lambda^2 \alpha^2$$

$$\Rightarrow m = \pm i\lambda$$

$$T = C_6 e^{-\lambda^2 \alpha^2 t}$$

$$x = C_4 \cos \lambda x + C_5 \sin \lambda x$$

Case (iii): when $k=0$

$$\frac{d^2x}{dx^2} = 0 \quad ; \quad \frac{dT}{dt} = 0$$

$$\Rightarrow \frac{dx}{dx} = C_7$$

$$dT = 0$$

$$T = C_9$$

$$\Rightarrow x = C_7 x + C_8$$

Therefore the possible solutions of the heat equations are

$$u(x,t) = (C_1 e^{\lambda x} + C_2 e^{-\lambda x}) C_3 e^{\alpha^2 \lambda^2 t}$$

$$u(x,t) = (C_4 \cos \lambda x + C_5 \sin \lambda x) C_6 e^{-\alpha^2 \lambda^2 t}$$

$$u(x,t) = (C_7 x + C_8) C_9$$

of these three solutions, we have to choose the solution which is consistent with the physical nature of the problem. Since u decreases as t increases, the only suitable solution of the heat equation is,

$$u(x,t) = (A \cos \lambda x + B \sin \lambda x) e^{-\alpha^2 \lambda^2 t}$$

Problems:-

Q. Solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ with boundary conditions $u(x, 0) = 3 \sin n\pi x$, $u(0, t) = 0$, $u(1, t) = 0$ where $0 < x < 1$, $t > 0$.

Soln:

The solution of the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \rightarrow (1)$ is $u(x, t) = (A \cos \lambda x + B \sin \lambda x) e^{-\lambda^2 t} \rightarrow (2)$

when $x = 0$,

$$u(0, t) = A e^{-\lambda^2 t} = 0 \Rightarrow A = 0.$$

$\therefore (2)$ becomes, $u(x, t) = B(\sin \lambda x) e^{-\lambda^2 t} \rightarrow (3)$

when $x = 1$,

$$u(1, t) = B(\sin \lambda) e^{-\lambda^2 t} = 0$$

$$\Rightarrow \sin \lambda = 0 \quad (B \text{ cannot be zero as } e^{-\lambda^2 t} \neq 0)$$

$$\Rightarrow \lambda = n\pi$$

$$\therefore (3) \text{ becomes, } u(x, t) = B(\sin n\pi x) e^{-n^2 \pi^2 t} \\ = B_n(\sin n\pi x) e^{-n^2 \pi^2 t}.$$

Thus the general solution of (1) is,

$$u(x, t) = \sum_{n=1}^{\infty} B_n e^{-n^2 \pi^2 t} \sin n\pi x \rightarrow (4)$$

$$\text{when } t = 0, \quad 3 \sin n\pi x = u(x, 0) = \sum_{n=1}^{\infty} B_n \sin n\pi x$$

comparing both sides $B_n = 3$,

Hence from (4), the desired solution is,

$$u(x,t) = 3 \sum_{n=1}^{\infty} \sin n\pi x e^{-n^2\pi^2 t}.$$

Q. Solve the differential $\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}$ for the conduction of heat along a rod without radiation, subject to the following conditions:

i) u is not infinite for $t \rightarrow \infty$, ii) $\frac{\partial u}{\partial x} = 0$ for $x=0$ and $x=l$, iii) $u = l x - x^2$ for $t=0$ between $x=0$ and $x=l$.

Soln:

Let $u = x T(t)$ be the trial solution.

Then $\frac{\partial u}{\partial t} = x T'$ and $\frac{\partial^2 u}{\partial x^2} = x'' T$

Thus $\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2} \Rightarrow x T' = \alpha^2 x'' T$

$$\Rightarrow \frac{x''}{x} = \frac{1}{\alpha^2} \frac{T'}{T} = \lambda^2 \text{ (say)}$$

Thus $\frac{x''}{x} = \lambda^2 \Rightarrow x'' - \lambda^2 x = 0 \longrightarrow (i)$

$$\Rightarrow x = c_1 e^{\lambda x} + c_2 e^{-\lambda x}$$

$$\frac{1}{\alpha^2} \frac{T'}{T} = \lambda^2 \Rightarrow T = c_3 e^{\lambda^2 \alpha^2 t}.$$

Thus $u(x,t) = (c_1 e^{\lambda x} + c_2 e^{-\lambda x}) (c_3 e^{\lambda^2 \alpha^2 t}) \longrightarrow (ii)$

Thus $u(x,t) = (c_1 e^{\lambda x} + c_2 e^{-\lambda x}) (c_3 e^{\lambda^2 \alpha^2 t}) \longrightarrow (1)$

If λ^2 is changed to $-\lambda^2$, we get the solution as

$$u(x,t) = (C_4 \cos \lambda x + C_5 \sin \lambda x) C_6 e^{-\lambda^2 x^2 t} \quad (2)$$

If $\lambda^2 = 0$, we get $u(x,t) = (C_7 x + C_8) C_9 \quad (3)$.

In (1), $T \rightarrow \infty$ for $t \rightarrow \infty$ therefore u also $\rightarrow \infty$.
the given condition (i) is not satisfied.
So we reject that solution (1), while

(2) and (3) satisfy the condition.

Apply the condition $\frac{\partial u}{\partial n} = 0$ to (3) we get $C_7 = 0$.

$$\therefore u = xT = C_8 C_9 = a_0 \text{ (say)} \quad (4)$$

$$\text{From (2)} \quad \frac{\partial u}{\partial n} = (-C_4 \sin \lambda x + C_5 \cos \lambda x) \lambda C_6 e^{-\lambda^2 x^2 t}$$

Applying the condition $\left(\frac{\partial u}{\partial n} = 0\right)_{n=0}$ we get

$$0 = C_5 \lambda C_6 e^{-\lambda^2 x^2 t}$$

$$\Rightarrow C_5 = 0$$

$$\therefore u(x,t) = (C_4 \cos \lambda x) C_6 e^{-\lambda^2 x^2 t}$$

$$\left(\frac{\partial u}{\partial n}\right)_{n=l} = 0 \Rightarrow 0 = (-C_4 \sin \lambda l + C_5 \cos \lambda l) \lambda C_6 e^{-\lambda^2 x^2 t}$$

we have $C_5 = 0$.

$$\therefore 0 = (-C_4 \sin \lambda l) \lambda C_6 e^{-\lambda^2 x^2 t}$$

$$\sin \lambda l = 0 \Rightarrow \lambda l = n\pi$$

$n\pi/l$

$$\therefore u = (c_4 \cos \lambda x) e^{-\lambda^2 x^2 t}$$

$$= B_n \cos \frac{n\pi x}{l} \cdot e^{-\frac{n^2 \pi^2 x^2 t}{l^2}} \quad (c_4 l = B_n)$$

→ (5)

Thus the general solution being the sum of (4) and (5) is

$$u = a_0 + \sum_{n=1}^{\infty} B_n \cos\left(\frac{n\pi x}{l}\right) e^{-\frac{n^2 \pi^2 x^2 t}{l^2}}$$

Now using (iii), we get,

$$lx - x^2 = a_0 + \sum_{n=1}^{\infty} B_n \cos\left(\frac{n\pi x}{l}\right)$$

This being the expansion of $lx - x^2$ as a half range cosine series in $(0, l)$, we get

$$a_0 = \frac{1}{l} \int_0^l (lx - x^2) dx$$

$$= \frac{1}{l} \left[\frac{lx^2}{2} - \frac{x^3}{3} \right]_0^l = \frac{1}{l} \left[\frac{l^3}{2} - \frac{l^3}{3} \right]$$

$$= \underline{\underline{\frac{l^2}{6}}}$$

$$a_n = \frac{2}{l} \int_0^l (lx - x^2) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2}{l} \left[(lx - x^2) \left(\frac{l}{n\pi} \sin \frac{n\pi x}{l} \right) - (l - 2x) \left(\frac{-l^2}{n^2 \pi^2} \cos \frac{n\pi x}{l} \right) \right. \\ \left. + (-2) \left[\frac{-l^3}{n^3 \pi^3} \sin \left(\frac{n\pi x}{l} \right) \right] \right]_0^l$$

$$= \frac{2}{l} \left\{ 0 - \frac{l^3}{n^2 \pi^2} (\cos n\pi + 1) + 0 \right\} = \begin{cases} -\frac{4l^2}{n^2 \pi^2} & \text{when } n \\ & \text{is even} \\ 0, & \text{otherwise.} \end{cases}$$

Taking $n = 2m$, the required solution is

$$u = \frac{l^2}{6} - \frac{l^2}{\pi^2} \sum_{m=1}^{\infty} \frac{1}{m^2} \cos\left(\frac{2m\pi x}{l}\right) e^{-4m^2 \pi^2 x^2 t / l^2}$$

- a. An insulated rod of length l has its ends A and B maintained at 0°C and 100°C respectively until steady state conditions prevail. If B is suddenly reduced to 0°C and maintained at 0°C , find the temperature at a distance x from A at time t .
- b) Solve the above problem if the change consists of raising the temperature of A to 20°C and reducing that of B to 80°C .

Soln:

Let the equation for the conduction of heat be $\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}$ $\longrightarrow$ ①

Prior to the temperature change at the end B when $t=0$, the heat flow was independent of time (steady state condition) when u depends only on x , ① reduces to $\frac{\partial^2 u}{\partial x^2} = 0$.

Its general solution is $u = ax + b$. $\longrightarrow$ ②

Since $u = 0$ for $x = 0$ and $u = 100$ for $x = l$

$\therefore$ ② gives $b = 0$ and $a = \frac{100}{l}$

Thus the initial condition is expressed

$$\text{By } u(x, 0) = \frac{100}{l} x. \longrightarrow (3)$$

Also the boundary conditions for the subsequent flow are,

$$\left. \begin{aligned} u(0, t) &= 0 \text{ for all values of } t, \\ u(l, t) &= 0 \text{ for all values of } t. \end{aligned} \right\} \longrightarrow (4)$$

Thus we have to find a temperature function $u(x, t)$ satisfying the differential equation (1) subject to the initial condition $u(x, 0) = \frac{100x}{l}$ and the boundary conditions

$$\text{Now the solution of (1) is of the form} \\ u(x, t) = (A \cos \lambda x + B \sin \lambda x) e^{-\alpha^2 \lambda^2 t} \longrightarrow (5)$$

By $u(0, t) = 0$, (5) reduces to,

$$0 = A e^{-\alpha^2 \lambda^2 t}$$

$$\Rightarrow A = 0.$$

$$\text{Hence (5)} \Rightarrow u(x, t) = (B \sin \lambda x) e^{-\alpha^2 \lambda^2 t} \longrightarrow (6)$$

By $u(l, t) = 0$, (6) becomes

$$0 = (B \sin \lambda l) e^{-\alpha^2 \lambda^2 t}$$

$$\Rightarrow \sin \lambda l = 0$$

$$\Rightarrow \lambda l = n\pi$$

$$\Rightarrow \lambda = n\pi/l$$

$$\therefore \textcircled{6} \Rightarrow u(x,t) = B \sin\left(\frac{n\pi x}{l}\right) e^{-\frac{\alpha^2 n^2 \pi^2 t}{l^2}}$$

$$= B_n \sin\left(\frac{n\pi x}{l}\right) e^{-\frac{\alpha^2 n^2 \pi^2 t}{l^2}} \quad (B_n = B)$$

Adding all such equations, the most general solution of ① satisfying the boundary condition ④ is

$$u(x,t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{l}\right) e^{-\frac{\alpha^2 n^2 \pi^2 t}{l^2}} \longrightarrow \textcircled{7}$$

Putting $t = 0$, $u(x,0) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{l}\right) \longrightarrow \textcircled{8}$

In order that the condition ③ may be satisfied

③ and ⑧ must be same. This requires the expansion of $\frac{100x}{l}$ as a half range Fourier sine series in $(0, l)$. Thus,

$$\frac{100x}{l} = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{l} \quad \text{where}$$

$$B_n = \frac{2}{l} \int_0^l \frac{100x}{l} \sin \frac{n\pi x}{l} dx$$

$$= \frac{200}{l^2} \left[x \left(-\frac{\cos \frac{n\pi x}{l}}{\frac{n\pi}{l}} \right) - \left(-\frac{\sin \frac{n\pi x}{l}}{\left(\frac{n\pi}{l} \right)^2} \right) \right]_0^l$$

$$= \frac{200}{l^2} \left[-l^2 \frac{\cos n\pi}{n\pi} \right]$$

$$= -\frac{200}{n\pi} (-1)^n = \frac{200}{n\pi} (-1)^{n+1}$$

Hence (7) gives, $u(x,t) = \frac{200}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin \frac{n\pi x}{l} e^{-\frac{n^2 \pi^2 t}{l^2}}$

b) Here the initial condition remains the same as (3) above and the boundary conditions are $u(0,t) = 20$, $u(l,t) = 80$. $\longrightarrow$ (9)

In part a) the boundary values being zero, we are able to find the desired solution easily. Now the boundary values being non-zero, we have to modify the procedure. We split up the temperature function $u(x,t)$ in to two parts as,

$$u(x,t) = u_s(x) + u_t(x,t) \longrightarrow (10)$$

where $u_s(x)$ is a solution of (1) involving x only and satisfying the boundary conditions (9). $u_t(x,t)$ is then a function defined by (10).

Thus $u_s(x)$ is a steady state solution of the form (2) involving x only and satisfying the boundary conditions (9). $u_t(x,t)$ is then a function defined by (10).

Since $u_s(0) = 20$ and $u_s(l) = 80$

$\therefore$ using (2), we get

$$u_s(x) = 20 + \left(\frac{60}{l}\right)x$$

Putting $x=0$ in (10) we have,

$$u_f(0,t) = u(0,t) - u_s(0) = 20 - 20 = 0. \longrightarrow (11)$$

Putting $x=l$, we have

$$u_f(l,t) = u(l,t) - u_s(l) = 80 - 80 = 0. \longrightarrow (12)$$

$$\begin{aligned} u_f(x,0) &= u(x,0) - u_s(x) = \frac{100x}{l} - \left(\frac{60x}{l} + 20\right) \\ &= \frac{40x}{l} - 20. \longrightarrow (13) \end{aligned}$$

Hence (11) and (12) give the boundary conditions and (13) gives the initial condition relative to the transient solution.

$$\text{Thus } u_f(x,t) = (A \cos \lambda x + B \sin \lambda x) e^{-\lambda^2 a^2 t}. \longrightarrow (14)$$

$$\text{By (12), } u_f(0,t) = A e^{-\lambda^2 a^2 t} = 0.$$

$$\text{Hence } A=0. \Rightarrow u_f(x,t) = B \sin \lambda x e^{-\lambda^2 a^2 t} \longrightarrow (15)$$

$$u_f(l,t) = 0 \Rightarrow u_f(l,t) = B \sin \lambda l e^{-\lambda^2 a^2 t} = 0$$

$$\Rightarrow \lambda l = n\pi \Rightarrow \lambda = \frac{n\pi}{l}$$

$$\text{Hence (15) } \Rightarrow u_f(x,t) = B_n \sin \frac{n\pi x}{l} e^{-\frac{n^2 \pi^2 a^2 t}{l^2}} \text{ where } B=B_n.$$

Adding all such solutions,

$$u_f(x,t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{l}\right) e^{-\frac{n^2 \pi^2 a^2 t}{l^2}} \longrightarrow (16)$$

$$\text{Putting } t=0, \text{ we have } u_f(x,0) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{l} \longrightarrow (17)$$

In order that the condition (13) may be satisfied, (13) and (17) must be the same

This requires the expansion of $(\frac{40}{l})x - 20$ as a half range Fourier sine series in $(0, l)$. Thus

$$\frac{40x}{l} - 20 = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{l} \quad \text{where}$$

$$B_n = \frac{2}{l} \int_0^l \left(\frac{40x}{l} - 20 \right) \sin \left(\frac{n\pi x}{l} \right) dx$$

$$= -\frac{A_0}{n\pi} (1 + \cos n\pi).$$

$B_n = 0$ when n is odd.

$$B_n = -\frac{80}{n\pi}, \text{ when } n \text{ is even}$$

Hence (16) becomes,

$$u_t(x, t) = \sum_{n=2,4,\dots}^{\infty} \left(-\frac{80}{n\pi} \right) \sin \left(\frac{n\pi x}{l} \right) e^{-a^2 n^2 \pi^2 t / l^2} \quad \text{--- (18)}$$

$$x = \frac{40}{l}x$$

Finally combining $u_s(x)$ and $u_t(x, t)$, the required solution is,

$$u(x, t) = \left[20 + \frac{60x}{l} \right] + \sum_{n=2,4,\dots}^{\infty} \frac{-80}{n\pi} \sin \left(\frac{n\pi x}{l} \right) e^{-a^2 n^2 \pi^2 t / l^2}$$

- Q. A bar 100 cm long with insulated sides, has its ends kept at 0°C and 100°C until steady state conditions prevail. The two ends are then suddenly insulated and kept so. Find the temperature distribution.

Soln: The temperature $u(x, t)$ along the bar

satisfies the equation $\frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2}$ $\longrightarrow$ ①

By law of heat conduction, the rate of heat flow is proportional to the gradient of the temperature

Thus if the ends $x=0$ and $x=l$ of the bar are insulated, so that no heat can flow through the ends. The boundary conditions are,

$$\frac{\partial u}{\partial x}(0, t) = 0, \quad \frac{\partial u}{\partial x}(l, t) = 0 \quad \text{for all } t. \longrightarrow ②$$

Initially, under steady state conditions,

$$\frac{\partial^2 u}{\partial x^2} = 0. \quad \text{Its solution is } u = ax + b.$$

Since $u = 0$ for $x = 0$ and $u = 100$ for $x = l$.
 $\therefore b = 0$ and $a = l$.

Thus initial condition is $u(x, 0) = x, 0 < x < l$ $\longrightarrow$ ③

Now the solution of ① is of the

$$\text{form } u(x, t) = (A \cos \lambda x + B \sin \lambda x) e^{-a^2 \lambda^2 t}. \longrightarrow ④$$

D.p. w.r.t. x ,

$$\frac{\partial u}{\partial x} = (-A \lambda \sin \lambda x + B \lambda \cos \lambda x) e^{-a^2 \lambda^2 t}. \longrightarrow ⑤$$

$$\text{Putting } x=0, \left(\frac{\partial u}{\partial x}\right)_0 = B \lambda e^{-a^2 \lambda^2 t} = 0 \quad \text{for all } t.$$

$$\text{Putting } x=l \text{ in } ⑤ \left(\frac{\partial u}{\partial x}\right)_l = -B \lambda \sin \lambda l e^{-a^2 \lambda^2 t} = 0 \quad \text{for all } t.$$

$$\therefore A \lambda \sin \lambda l = 0 \text{ is } \lambda \text{ being } \neq 0 \text{ either } A = 0 \text{ or } \sin \lambda l = 0.$$

$$\therefore l = n\pi \Rightarrow \lambda = n\pi/l$$

Hence (4) becomes, $u(x,t) = A \cos \frac{n\pi x}{l} e^{-a^2 n^2 \pi^2 t / l^2}$

$\therefore$ The most general solution of (1) satisfying the boundary conditions (2) is,

$$u(x,t) = \sum_{n=0}^{\infty} B_n \cos\left(\frac{n\pi x}{l}\right) e^{-a^2 n^2 \pi^2 t / l^2} = B_0 + \sum_{n=1}^{\infty} B_n \cos \frac{n\pi x}{l} e^{-a^2 n^2 \pi^2 t / l^2} \quad (6)$$

putting $t=0$, $u(x,0) = B_0 + \sum_{n=1}^{\infty} B_n \cos \frac{n\pi x}{l} = x$

This requires the expansion of x into a half-range cosine series in $(0,l)$.

$$\text{Thus } x = \frac{a_0}{2} + \sum_{n=1}^{\infty} B_n \cos \frac{n\pi x}{l}$$

$$\text{where } a_0 = \frac{2}{l} \int_0^l x \, dx = l$$

$$\text{and } a_n = \frac{2}{l} \int_0^l x \cos \frac{n\pi x}{l} \, dx$$

$$= \frac{2l}{n^2 \pi^2} (\cos n\pi - 1)$$

$$= 0, \text{ where } n \text{ is even}$$

$$= \begin{cases} -\frac{4l}{n^2 \pi^2} & \text{for } n \text{ is odd.} \end{cases}$$

$$B_0 = \frac{a_0}{2} = l/2 \text{ and } B_n = a_n = 0 \text{ for } n \text{ is even}$$

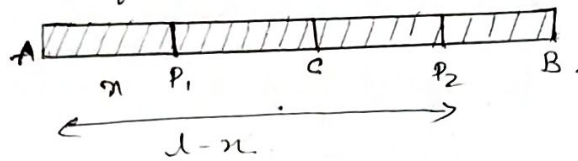
$$= -\frac{4l}{n^2 \pi^2} \text{ for } n \text{ is odd}$$

Hence (6) takes the form

$$u(x,t) = l/2 + \sum_{n=1,3,\dots}^{\infty} \frac{4l}{n^2 \pi^2} \cos \frac{n\pi x}{l} e^{-a^2 n^2 \pi^2 t / l^2}$$

$$= l/2 - \frac{4l}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos \frac{(2n-1)\pi x}{l} e^{-a^2 (2n-1)^2 \pi^2 t / l^2} \quad (7)$$

This is the required temperature at a point P , distant x from end A at any time t .



Let P_1, P_2 be two points equidistant from the centre C of the bar so that $CP_1 = CP_2$

If $AP_1 = BP_2 = x$ then $AP_2 = l-x$

$\therefore$ Replacing x by $l-x$ in (7) we get the

temperature at P_2 as,

$$u(l-x, t) = \frac{1}{2} - \frac{4l}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos \frac{(2n-1)\pi(l-x)}{l} e^{-\frac{a^2(2n-1)^2\pi^2 t}{l^2}}$$

$$= \frac{1}{2} + \frac{4l}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos \frac{(2n-1)\pi x}{l} \quad \text{--- (8)}$$

$$\because \cos \frac{(2n-1)\pi(l-x)}{l} = \cos \left[2n\pi - \pi - \left(\frac{(2n-1)\pi x}{l} \right) \right]$$

$$= -\cos \frac{(2n-1)\pi x}{l}$$

Adding (7) and (8), we get $u(x, t) + u(l-x, t) = 100^\circ\text{C}$

Question Bank

- 1) A tightly stretched flexible string has its ends fixed at $x=0$ and $x=l$. At time $t=0$, the string is given a shape defined by $F(x) = \mu x(l-x)$ where μ is a constant, and then released. Find the displacement of any point x of the string at any time $t > 0$.
- 2) A tightly stretched string of length l with fixed endpoints is initially in equilibrium position. It is set vibrating by giving each point a velocity $v_0 \sin^3 \pi x/l$. Find the displacement $y(x,t)$.
- 3) The points of trisection of a string are pulled aside through the same distance on opposite sides of the position of equilibrium and the string is released from rest. Derive an expression for the displacement of the string at subsequent time and show that the midpoint of the string always remains at rest.
- 4) The ends A and B of a rod 20 cm long have temperature at 30°C and 80°C until steady state conditions prevail. The temperature of the ends are changed to 40°C and 60°C respectively. Find the temperature distribution in the rod at time t .

- 5) Find the temperature in a laterally insulated bar of length L whose ends are kept at temperature zero if the initial temperature is

$$f(x) = \begin{cases} x, & 0 < x < L/2 \\ L-x, & L/2 < x < L \end{cases}$$

- 6) An insulated rod of length L has its ends A and B maintained at 0°C and 100°C respectively until steady state conditions prevail. If B is suddenly reduced to 0°C and maintained at 0°C then find the temperature on the rod at a distance x from A at time t .